

MLFF-DATA9A9b Production DATA6-DATA8 Materialization

mdstats project

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1. Purpose

DATA9A9b closes the orchestration gap between a completed checkpoint-bound DATA6 model sweep and executable final/fold DATA8 training jobs. DATA7 and DATA8 scientific semantics already exist. This stage makes their production realization exact, restartable, tamper-evident, and bound to one replay corpus.

The stage owns neither feature definitions nor MACE training. It owns:

- freezing the complete DATA6 sweep identity;
- freezing every canonical final and cross-validation DATA7 domain;
- freezing all DATA7 fitting, objective, weighting, checkpoint, and selection policies;
- binding one exact replay-train/replay-monitor plan;
- checkpointing each realized DATA7 bundle independently;
- invalidating downstream DATA8 evidence when any DATA7 artifact changes;
- promoting a complete final/fold DATA8 tree only after all prerequisites pass;
- emitting one immutable materialization record that can update production qualification.

DATA9B remains closed until a complete production DATA9A9b record exists.

2. Ownership boundary

`mdstats.training_data.production_materialization` owns orchestration and restart evidence only.

- DATA6 owns checkpoint-bound descriptors, predictions, residuals, and blinded summaries.
- DATA7 owns fold-local feature fitting, E0 fitting, objectives, weights, selection, and coverage.
- DATA8 owns extxyz export, MACE configuration, replay staging, loader dry runs, and sealed evaluation metadata.
- Replay owns replay-file inspection, disjointness, label provenance, and retention policy.
- DATA9B owns training execution, checkpoint comparison, committee construction, and freeze.

The production orchestrator calls these owners and records their native bundle digests. It does not reproduce their algorithms.

3. Frozen production plan

`ProductionMaterializationPlan` binds:

- source, frame, DATA4, DATA5, and DATA6 identities;
- the complete DATA6 model-sweep checkpoint;
- exact descriptor and prediction manifests;
- every canonical `FeatureFitDomain` from DATA5;
- feature-metric, atomic-reference, objective, configuration-weight, checkpoint-metric, and selection-budget policies;
- the exact foundation checkpoint;
- optional foundation atomic reference energies;
- MACE compatibility policy and passing source probe;

- the exact replay train and monitor artifacts;
- optimizer, checkpoint-control, and extxyz policies;
- selected DATA7 ladder size and replay exposure settings.

The canonical domain set contains exactly one final-development domain and one training domain for each declared cross-validation fold. The plan is rejected when the replay plan is unresolved, when the completed DATA6 sweep differs from the DATA6 bundle, or when the DATA8 foundation checkpoint differs from the checkpoint used for DATA6 predictions.

4. Restart semantics

Each DATA7 domain is written independently to `data7/<domain_digest>.json`. `ProductionData7ArtifactRecord` binds:

- domain digest;
- DATA7 bundle digest;
- safe relative path;
- file SHA-256.

`ProductionMaterializationCheckpoint` is atomically replaced after every newly completed domain. A bounded run may stop after `max_new_data7_domains`; a later run verifies and reuses all valid records.

If a DATA7 artifact is missing, malformed, modified, or bound to a foreign DATA6 bundle, it is either rejected or recomputed according to the explicit execution policy. Any invalid DATA7 artifact invalidates the DATA8 tree because the training protocol depends on its fitted transforms, E0 values, selection, and weights.

5. Exact replay binding

Production multihead replay requires two disjoint files:

- replay training data;
- replay monitor data.

The plan binds their configuration counts, atomic-number sets, geometry identities, numerical energy/force/stress label identities, label keys, label provenance, foundation-checkpoint identity for pseudo-labels, selection provenance, head weight, target weight, and retention policy.

DATA8 may stage and rescale the replay-train file to realize fixed-file MACE weights. Therefore source and staged file byte hashes need not be equal. The orchestrator verifies semantic equivalence: configuration identity, labels, provenance, counts, species, and replay policy must remain unchanged.

6. DATA8 promotion

DATA8 construction begins only when every planned DATA7 domain is valid. It is built in hidden staging, verified, moved into a content-addressed generation directory, and exposed by an atomic `data8` symlink switch. The output tree contains:

- one final-development job;
- one job per cross-validation fold;
- selected target-train and target-monitor extxyz files;
- fold evaluation files that are never referenced by training configs;
- staged foundation and replay resources;
- MACE configs, commands, job manifests, and loader dry-run evidence;
- sealed locked-test metadata without materialized locked-test labels.

A DATA8 artifact record binds the native DATA8 bundle digest and a deterministic relative-path/file-SHA tree manifest. Completion is promoted only after the whole tree and bundle can be reloaded and verified. Failed construction removes an unpromoted partial DATA8 directory and records failure evidence.

7. Atomic-reference modes

For foundation-model fine-tuning, production normally uses `FOUNDATION_RESIDUAL`. In that mode every fit-domain frame requires a persisted foundation energy prediction, the exact foundation checkpoint digest, and a complete foundation E0 mapping for all target elements.

From-scratch total-energy E0 fitting remains available for explicit tests and non-transfer workflows. Foundation predictions or E0s must not be supplied to a from-scratch fit.

8. Materialization record

`ProductionMaterializationRecord` exposes:

- plan and checkpoint identity;
- ordered DATA7 bundle digests;
- DATA8 bundle digest;
- completion state;
- verified loading of native DATA7 and DATA8 bundles.

The filesystem root is a relocatable location hint and is excluded from the scientific content digest. The record supplies the exact DATA7/DATA8 digests and replay binding needed by `ProductionCorpusQualificationRecord`. The production qualification builder accepts the complete record directly, verifies DATA6 lineage, loads the native bundles, and derives replay/materialization status without caller-supplied duplicate digests.

9. Failure rules

The stage fails closed for:

- incomplete or foreign DATA6 model-sweep evidence;
- changed source, frame, DATA4, DATA5, or DATA6 identity;
- a foundation checkpoint mismatch;
- unresolved or absent replay when replay is required;
- foreign or noncanonical DATA7 domains;
- modified DATA7 files;
- modified DATA8 files or tree membership;
- missing MACE compatibility support;
- selection sizes absent from a DATA7 ladder;
- replay/target exact geometry overlap;
- incomplete residual-E0 evidence;
- any attempt to materialize DATA8 before all DATA7 domains pass.

10. Focused acceptance tests

The release gate must cover:

1. bounded one-domain execution followed by deterministic resume;
2. one final and all fold-local DATA7 bundles;
3. exact replay binding in every DATA8 protocol;
4. no evaluation or locked-test file in a MACE training configuration;
5. DATA7 tamper detection and downstream DATA8 invalidation;
6. DATA8 tree tamper detection;
7. plan mismatch rejection;
8. historical DATA6 compatibility remaining unchanged;
9. source/wheel export and registry parity;
10. immutable record serialization and relocation-safe identity.

11. Scope of the current implementation

Version 0.20.55a0 retains and hardens the complete restartable control layer and qualifies it on a bounded four-domain target/replay workflow. It does not claim that the full 2,734-frame production sweep has completed in this execution environment. The production corpus must run DATA9A9a to completion, then execute this DATA9A9b plan before DATA9B training begins.